Spectral Variability-Aware Cascaded Autoencoder for Hyperspectral Unmixing

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***Abstract*— Spectral variability inevitably presents in hyperspectral images (HSIs), resulting in significant unmixing errors when using the conventional linear mixture model (LMM). Though several variants of LMM have been proposed to encounter such spectral variability, they cannot well model the complex characteristics of spectral variability, and the performance of these variants strongly depends on the prior knowledge of the scene. In this article, spectral variability within an image is classified into class-dependent variability and class-independent one, which can be tackled by a novel fully linear mixture model (FLMM) introducing a class-dependent multiplicative scaling term, a class-dependent additive perturbation term, and a class-independent variability term into the conventional LMM. Moreover, a spectral variability-aware cascaded autoencoder (SVACA) is designed to realize the automatic learning and representation of unmixing targets and spectral variability in different hyperspectral scenarios, which consists of a class-independent variability autoencoder and a cascaded class-dependent variability autoencoder. Such a network is able to handle different spectral variability autonomously without any scene prior by parallel inference structure. Experimental results over synthetic and real hyperspectral datasets demonstrate that the proposed SVACA network not only outperforms several state-of-the-art unmixing networks but also presents a stronger capability to handle spectral variability within HSIs.**

***Index Terms*— Autoencoder, deep learning, hyperspectral images (HSIs), spectral variability.**

I. Introduction

HYPERSPECTRAL images (HSIs), owing to the properties of high spectral resolution and image-spectrum unification, have shown the powerful ability to identify and distinguish ground targets. Therefore, HSIs have been widely applied in civil and military fields [1], [2], [3]. However, given the relatively low spatial resolution of the imaging system and the complex distribution of ground objects, the phenomenon of mixed pixels commonly exists in HSIs, which significantly limits the quantitative analysis of imaged scenes [4], [5]. Therefore, hyperspectral unmixing has been developed to decompose mixed pixels into a set of pure spectra (known as endmembers) and the fractional abundances (known as abundances), which is able to exploit subpixel information in HSIs [6], [7], [8].

Linear and nonlinear spectral unmixing are two major categories for the existing unmixing methods depending on different mixing models. Due to the computational tractability, linear mixture models (LMMs) have found extensive applications. Existing LMM-based unmixing methods can be geometrical [9], [10], statistics [11], [12], [13], sparse regression [14], [15], [16], [17], and deep-learning methods [18], [19], [20], [21], where endmember extraction and abundance estimation are termed as two key steps.

However, variable illumination, complex environmental conditions, and even different growth stages often cause inevitable variability of material spectra, i.e., spectral variability [22], [23], [24]. Such a phenomenon is commonly observed, which leads to the fact that pixels from the same pure material exhibit different spectral characteristics throughout the entire image. In this case, using LMM to identify one spectrum for each material will result in serious estimation errors during the unmixing process.

In the last decade, much effort has been dedicated to this problem and several parametric mixture models have been proposed [25], [26], [27], [28], [29], [30], [31]. Thouvenin et al. [32] modeled spectral variability by a perturbed linear mixing model (PLMM). In this model, each endmember's variability is represented by an additive perturbation term. However, the absence of any physical significance is one flaw in this model. In contrast, an extended linear mixing mode (ELMM), has been proposed in [33] and further generalized by [34]. This work mainly expressed spectral variability by scaling factors on endmembers. However, given that innegligible spectrum fluctuation (such as atmospheric effects or nonlinear spectral mixing) cannot be fully accounted for by scaling factors alone, this approach is unsatisfactory. More recently, Hong et al. [35] proposed an augmented linear mixing model (ALMM), which considered scaling factors and other spectral variability separately using endmember and spectral variability dictionaries. Zhang et al. [36] further worked on the

Received 25 August 2024; revised 5 December 2024 and 13 January 2025; accepted 10 February 2025. Date of publication 19 February 2025; date of current version 28 February 2025. This work was supported by the National Natural Science Foundation of China under Grant 62171381. *(Corresponding author: Shaohui Mei.)*

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Digital Object Identifier 10.1109/TGRS.2025.3543566

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extraction methods of spectral variability to enhance sparse unmixing performance by augmented linear sparse unmixing model.

However, the above parametric models are dependent on prior knowledge of the image scene under study, and they cannot effectively deal with spectral variability without prior information. Meanwhile, such models often take only one particular type of spectral variability into account, which causes limited applicability in different scenarios. Therefore, in this article, a novel fully linear mixture model (FLMM) is proposed to present a broadly applicable integrated and degenerable model to handle spectral variability in HSIs. This model simultaneously considered the two types of spectral variability: class-independent and class-dependent spectral variability, where class-dependent spectral variability is subdivided into additive perturbation and multiplicative scaling terms. Furthermore, a spectral variability-aware cascaded autoencoder (SVACA) is designed to implement the proposed model. Such a network realizes the whole process of spectral variability, endmember, and abundance estimation by the ingenious design of two cascaded autoencoders and residual structures. Therefore, the key contributions of this article can be summed up as follows.

1) The spectral variability within an HSI is separated into class-independent variability and class-dependent variability to improve the conventional LMM. Specifically, the class-dependent variability is further modeled as multiplicative scaling-form variability and additive perturbation-form variability to formulate a FLMM for mixed pixels. Such proposed FLMM can well model different kinds of spectral variability presented in an image and can degrade into many existing variants of LMM incorporating spectral variation, such as PLMM [32], ELMM [33], and generalized LMM (GLMM) [34].
2) An SVACA network is designed for the proposed FLMM, which consists of a class-independent variability autoencoder and a cascaded class-dependent variability autoencoder to well incorporate different kinds of spectral variability for linear unmixing. Specifically, the former autoencoder is responsible for modeling the class-independent varying spectrum caused by the interference by the atmospheric effects and the latter is in charge of the class-dependent varying spectrum characteristic caused by the scene's geometry and lighting conditions or the intrinsic change of matter.
3) In order to well incorporate both the multiplicative scaling-form and additive perturbation-form of class-dependent variability, the class-dependent variability autoencoder not only adopts parallel inference structure to learn abundance and scaling matrix simultaneously but also adds additive perturbation of residual-connected structure to the decoder to solve the problem of insufficient learning ability of single linear layer.

The remainder of this article is structured as follows. Brief reviews of related works, such as LMM and autoencoder variants, are included in Section II. Section III proposes a novel FLMM accounting for spectral variability. Section IV introduces the proposed SVACA architecture in detail. The experiments over synthetic and real datasets are reported in Section V. Section VI draws the conclusion.

II. Related Works

A. Linear Mixture Model

Let $\mathbf{R} \in \mathbb{R}^{b\times n}$ represent an observed HSI, owing b bands and n pixels. The LMM formulates the HSI as

$$\mathbf{R} = \mathbf{M}\mathbf{A} + \mathbf{N} \quad \text{s.t. } \mathbf{A} \geq 0, \mathbf{1}_P^T\mathbf{A} = \mathbf{1}_n^T \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{b\times P}$ denotes an endmember matrix consisting of P endmembers, $\mathbf{A} \in \mathbb{R}^{P\times n}$ represents the corresponding fractional abundances, and the error matrix is contained in $\mathbf{N} \in \mathbb{R}^{b\times n}$, including additive noise.

In pixel-wise, the LMM model can be expressed as

$$r_k = \sum_{p=1}^{P} a_{kp} m_p + n_k = \mathbf{M}a_k + n_k \tag{2}$$

where r_k denotes the kth pixel in the HSI, a_{kp} denotes the proportion of the kth pixel composed of the pth endmember, a_k denotes the abundance vector of the kth pixel, m_p denotes the pth endmember, and n_k denotes the additive noise contained in the kth pixel.

B. LMM Variants

1) Perturbed LMM: The PLMM [32] modeled each pixel by the endmember matrix affected by additional perturbation matrix $\mathbf{dm}_k \in \mathbb{R}^{b\times P}$ that account for its variability. Such perturbation allows pixel-specific variability. The PLMM can be written as

$$r_k = \sum_{p=1}^{P} a_{kp}(m_p + dm_{k,p}) + n_k = \mathbf{M}a_k + \mathbf{dm}_k a_k + n_k. \tag{3}$$

This model applies an additional perturbation matrix to the endmember variability, which explains the deviation from the LMM model in HSIs to a certain extent but lacks physical significance.

2) Extended LMM: By multiplying the endmember-wise scaling factor ψ_{kp} at each pixel, the ELMM [33] introduces a diagonal matrix to simulate the spectral variability. This enables pixel-wise variability of each endmember induced by the scene's geometry and lighting conditions. The ELMM can be written as

$$r_k = \sum_{p=1}^{P} a_{kp}\psi_{kp} m_p + n_k = \mathbf{M}\psi_k a_k + n_k. \tag{4}$$

Note that this model assumes that spectral characteristics under different wavelengths share the same changeable scale, which leads to a lack of flexibility especially when encountering more complex spectral distortion. Experimental studies of plant spectra under various environmental circumstances have revealed that the spectral variability is significantly dependent on the wavelength interval, however, the ELMM model cannot handle this type of spectral variability [37].

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3) Generalized LMM: The GLMM [34] further extends the ELMM to a generalized form, which allows band-dependent scaling factors to accommodate variability that may appear inconsistent in each band of the endmember spectra. The GLMM can be written as

$$r_k = \sum_{p=1}^{P} a_{kp}\left(\psi_{kp} \odot \mathrm{m}_p\right) + n_k = a_k(\psi_k \odot \mathbf{M}) + n_k \tag{5}$$

where $\odot$ represents the Hadamard product, which multiplies two elements in the same position separately.

This model is more flexible than the ELMM model and can be used for any particle size variability in each wavelength range, greatly enhancing the representation of spectral variability of the model. However, only using the scale factors alone is not sufficient due to its poor capacity to reflect atmospheric phenomena or nonlinear spectrum mixing.

4) Augmented LMM: On the basis of ELMM, the ALMM [35] introduces a separate term to represent additional spectral variability that cannot be described using pixel-wise scaling, which can be written as

$$r_k = \sum_{p=1}^{P} a_{kp}\psi_{kp}m_p + \sum_{n=1}^{N} e_n b_{kn} + n_k = \mathbf{M}\psi_k a_k + \mathbf{E}b_k + n_k \tag{6}$$

where $\mathbf{M}$ represents the extracted endmember dictionary in advance, $\mathbf{E} \in \mathbb{R}^{b\times P}$ denotes the unknown spectral variability dictionary, and b_k contains the representation coefficients corresponding to $\mathbf{E}$.

Although ALMM introduces additional spectral variability terms to obtain a more flexible representation, this model ignores the potential relationship between spectral variability and endmembers, which defines the spectral variability as being completely independent of endmembers.

C. Autoencoder Variants

Recently, deep learning-based methods have been successfully applied in computer vision, remote sensing image processing, and other fields for their superior performance [38], [39], where autoencoders as the benchmark network for unmixing have also received widespread attention. Specifically, autoencoder-based networks for hyperspectral unmixing tasks have roughly the following four variants: nonnegative sparse autoencoder [40], [41], [42], [43], variational autoencoder (VAE) [44], [45], adversarial autoencoder [46], [47], and convolutional autoencoder [48], [49], [50], [51], [52], [53], [54].

1) Nonnegative Sparse Autoencoder: Nonnegative sparse autoencoders have achieved excellent unmixing performance attributed to superior self-adaption abilities. Su et al. [40] proposed stacked nonnegative sparse autoencoders (SNSAs) for robust unmixing, which helps with outliers and low signal-to-noise ratio (SNR) issues. Ozkan et al. [41] proposed a two-stage sparse autoencoder to enable a set of endmember intrinsic properties like similarity and sparsity but bring the challenging problem of parameter selection. Qu and Qi [42] proposed an untied denoising autoencoder with sparsity (uDAS), in which the denoising constraints are further incorporated into the network optimization.

2) Variational Autoencoder: To solve the problem of hidden space irregularity, the VAE has been introduced to enforce regularization training during unmixing. Su et al. [44] proposed a deep autoencoder network (DAEN), which combines a stacked autoencoder and a VAE together to significantly enhance the robustness of the network. Borsoi et al. [45] proposed a deep generative unmixing algorithm (DeepGUn), which is able to learn the potential representation of each ground object in the scene and does not require a careful choice of regularization parameters.

3) Adversarial Autoencoder: Inspired by generative models, Jin et al. [46] proposed an unsupervised unmixing network based on adversarial autoencoder (AAENet), which employs adversative training to transfer the spatial relationship between nearby pixels. Min et al. [47] proposed a joint metric unmixing network (JMnet), which employs Wasserstein distance as a regularization term in the autoencoder and then performs feature matching in the discriminator.

4) Convolutional Autoencoder: By partially or completely replacing the fully connected layer in the autoencoders with a convolutional layer, the resulting convolutional autoencoders can effectively exploit the spatial correlation existing in HSIs and solve the deficiency of the traditional autoencoder in the utilization of spatial information. For instance, Zhang et al. [48] proposed convolutional autoencoders with inputs as groups or 3-D blocks to further merge spatial information. Palsson et al. [49] applied parallel convolutional autoencoders to spectral blocks to use spatial information for unmixing. Hua et al. [50] proposed an auto-encoder unmixing network incorporating spatial correlations between pixels during network training. Gao et al. [51] proposed a cycle-consistency unmixing network (CyCU-Net), which adopts two cascaded and cyclically executed convolutional autoencoders to merge high-level semantic information.

III. Problem Formulation

Recent studies indicated the spectral variability in HSIs mainly caused by atmospheric effects, illumination and topographic changes, and intrinsic variation of the spectral signatures of the materials [22], [24]. According to some prior assumptions, LMM and its variants (i.e., PLMM, ELMM, and GLMM), respectively, give corresponding parametric spectral mixing models. For instance, ELMM [33] has verified the multiplicative scaling can effectively simulate the change of observed reflectance caused by the scene's geometry and lighting conditions. PLMM [32] further clarified the change of characteristic spectrum caused by the intrinsic change of matter in which could not be represented by the multiplicative term can be solved by the additive perturbation. However, in practical situations, they cannot effectively deal with spectral variability without prior information. Meanwhile, such models mostly take only one particular type of spectral variability into account, which causes limited applicability in different scenarios.

In order to overcome the above shortcomings of existing models, in this section, a FLMM considering spectral

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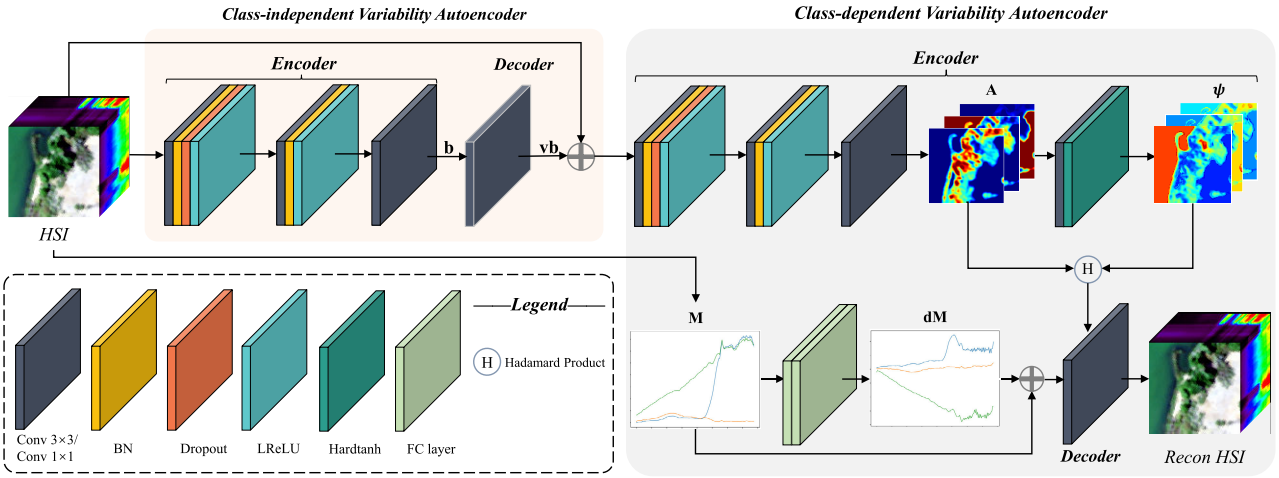


Fig. 1. Overall framework of the proposed SVACA, which consists of a class-independent variability autoencoder and a class-dependent variability autoencoder.

variability is proposed, which regards spectral variability as class-dependent and class-independent components. Specifically, the class-dependent spectral variability term reflects the differences in reflection characteristics of different classes of ground objects, corresponding to the spectral variability of different manifestations on different endmember spectra. Such a term is further divided into two parts: the observed reflectance change caused by the scene's geometry and lighting conditions is expressed as the class-dependent multiplicative scaling term; that of characteristic spectrum caused by the intrinsic change of matter is expressed as the class-dependent additive perturbation term.

Moreover, the interference by the atmospheric effects mainly reflects in the absorption of radiation by gases and aerosol and the scattering of incoming light by other molecules. The resulting changeable ground reflectance is independent of material characteristics so that the FLMM further introduces a class-independent spectral variability term devoted to solving the change of observed reflectance caused by atmospheric influence and other measurement errors. Note that such variability is manifested as random disturbances independent of the spectrum of ground objects and widely distributed throughout all pixels.

Therefore, the proposed FLMM can be formulated as

$$\begin{aligned} r_k &= \sum_{p=1}^{P} a_{kp}\psi_{kp}\left(m_p + dm_p\right) + \mathbf{V}b_k + n_k \\ &= \underbrace{(\mathbf{M}+\mathbf{dM})(\psi_k \odot a_k)}_{\text{class-dependent}} + \underbrace{\mathbf{V}b_k}_{\text{class-independent}} + n_k \end{aligned} \tag{7}$$

where $(\mathbf{M}+\mathbf{dM})$ represents the additive perturbation term of endmembers, $\mathbf{dM} \in \mathbb{R}^{b\times P}$ corresponds to the class-dependent perturbation variability on each endmember, $(\psi_k \odot a_k)$ denotes the multiplicative scaling variability on abundance, the vector ψ_k in scaling coefficient matrix $\Psi \in \mathbb{R}^{P\times n}$ is the variability degree of abundance of different ground objects in each pixel, $\mathbf{V}b_k$ denotes the class-independent spectral variability term, $\mathbf{V} \in \mathbb{R}^{b\times P}$ corresponds to the class-independent spectral variability within the whole image, and b_k denotes the corresponding representation coefficient.

TABLE I
COMPARISON OF SEVERAL MIXTURE MODELS ACCOUNTING FOR SPECTRAL VARIABILITY

	PLMM	ELMM	GLMM	ALMM	**FLMM**
Model Formulation	Eq. (3)	Eq. (4)	Eq. (5)	Eq. (6)	Eq. (7)
Scaling Variability		√	√	√	√
Perturbation Variability	√				√
Class-dependent Variability				√	√

Specifically, in the proposed FLMM, the endmember matrix in LMM is replaced by the sum of a basic endmember matrix and a variable endmember matrix, while the original abundance matrix is expressed as the fractional abundances with scaling levels. Since the scaling matrix Ψ acts in the form of Hadamard product, the scaling of the endmember matrix in each pixel can be equivalent to shrinking or enlarging of abundances of each pixel. Meanwhile, FLMM further introduces an additional spectral variability term accounting for class-independent components, i.e., $\mathbf{Vb_k}$. Such spectral variability $\mathbf{V}$ implicitly exists in the whole image and is generally distributed throughout all pixels.

The comparison between the proposed FLMM and existing mixture models accounting for spectral variability is concluded in Table I. Note that, unlike PLMM, by adapting the ideas underpinning ELMM, FLMM offers explicit physical concerns for scaling factors while modeling another variability under reliable physical assumptions. On the other hand, different from ELMM, with the introduction of variable endmember matrices, FLMM expands on the characterization of spectral variability that cannot be accounted for by multiplicative scaling. Furthermore, compared to ALMM, the proposed FLMM refines the patterns of spectral variability and builds a separate representation. Finally, the FLMM model can be degenerated into other LMM variants. For instance, when all elements of the scaling matrix Ψ are set to 1, the PLMM model is formed.

IV. PROPOSED SVACA

The common unmixing-oriented autoencoders transfer the observed image to low-dimension abundances by the encoder and then learn linear or nonlinear representation to reconstruct the input HSI by the decoder, whose weights can be considered

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TABLE II
NETWORK CONFIGURATION OF CLASS-INDEPENDENT VARIABILITY AUTOENCODER IN THE PROPOSED SVACA NETWORK

	Layer	Kernel Size	Filter	Stride	Activate Function
Encoder	Conv1	3×3	128	1	LReLU
	Conv2	3×3	64	1	LReLU
	Conv3	3×3	P	1	/
Decoder	Conv4	1×1	b	1	/

as endmembers. Such a relatively fixed structure ignores the significant effect of spectral variability. Therefore, in this section, an SVACA is presented to handle the spectral variability in the proposed FLMM. The overall framework of SVACA is present in Fig. 1, which consists of a class-independent variability autoencoder and a cascaded class-dependent variability autoencoder.

A. Class-Independent Variability Autoencoder

A basic autoencoder is introduced here to deal with the class-independent spectral variability, which mainly results in the change of observed reflectance caused by atmospheric influence and other measurement errors. The architecture details are listed in Table II. Specifically, the encoder includes three convolution layers with the kernel size as 3×3 (denoted as Conv1–Conv3), two batch normalization layers, two Leaky RELU activate function layers, and a dropout layer. As for the decoder, only a single convolution layer (denoted as Conv4) is employed to restore the noise-free HSI. Note that combining the common unmixing autoencoders and the proposed FLMM, the output of the encoder, that is a coefficient matrix, corresponds to the representation coefficient of class-independent spectral variability in the observed image, and the weight parameters of the decoder are considered as the class-independent variability spectra. In addition, the weight vectors of the decoder are initialized orthogonally.

B. Class-Dependent Variability Autoencoder

Based on the proposed FLMM in (7), the class-dependent variability autoencoder is designed to address the learning and representation of class-dependent variability term, in which the encoder estimates the multiplicative scaling-form variability caused by the scene's geometry and lighting conditions and the decoder learns the additive perturbation-form variability caused by the intrinsic change of matter. Table III shows the network configuration in detail.

To make full use of the encoder's powerful feature extraction ability, based on the basic encoder structure in Table II, an extra branch is designed to simultaneously learn the scaling vectors in Ψ for abundances (denoted as ConvSkip). This branch includes a 1×1 convolution layer and a Hardtanh activation function layer to restrict the scaled value in the range of (0.5, 1.5). Then, the scaled abundances are acquired by multiplying the two by the Hadamard product. The Hardtanh function is defined as

$$\text{Hardtanh}(x) = \begin{cases} 1.5 & \text{if } x \geq 1.5 \\ 0.5 & \text{if } x \leq 0.5 \\ x & \text{others.} \end{cases} \tag{8}$$

Algorithm 1 Proposed SVACA Network

Input: Hyperspectral data matrix $\mathbf{R} \in \mathbb{R}^{b \times n}$; Parameters λ_1, λ_2, and λ_3.

Output: Endmembers $\mathbf{M} \in \mathbb{R}^{n \times P}$; Abundance fractions $\mathbf{A}$; the class-dependent multiplicative scaling variability Ψ; the class-dependent additive perturbation variability $\mathbf{dM}$; the class-independent spectral variability $\mathbf{V}$; the class-independent variability representation coefficient $\mathbf{b}$.

Initialize $\mathbf{M}$ by VCA and $\mathbf{V}$ orthogonally; $\Psi = 1$; $\mathbf{dM} = 0$. **Repeat**:
  Class-independent Variability Autoencoder:
    Calculate $\mathbf{b}_t = f_{encoder}(\mathbf{R})$;
    Calculate $\mathbf{V}\mathbf{b}_t = f_{decoder}(\mathbf{b}_t)$;
  Class-dependent Variability Autoencoder:
    Calculate $\mathbf{A}_t = g_{encoder}(\mathbf{R}_t - \mathbf{V}\mathbf{b}_t)$;
    Calculate $\Psi_t = \text{Hardtanh}(Conv_{1\times1}(\mathbf{A}_t))$;
    Calculate $\mathbf{dM}_t = \text{Linear}(\mathbf{M}_t)$;
    Calculate $\mathbf{M}_t = \mathbf{M}_t + \mathbf{dM}_t$;
  Calculate $\mathbf{R}'_t = g_{decoder}(\mathbf{A}_t \cdot \Psi_t)$;
  Calculate the total loss function value $\mathcal{L}_t$ as (9);
  $t \leftarrow t + 1$;
Until Convergence: $\frac{\|\mathcal{L}_{t+1} - \mathcal{L}_t\|}{\|\mathcal{L}_t\|} < 10^{-5}$.

As for the decoder, a novel initialization method is designed, that is, the residual structure is introduced to learn an additive perturbation for the endmember matrix through the two fully connected layers (denoted as FC1 and FC2), where the initial endmembers extracted by traditional endmember extraction algorithms is used as the input. Different from the commonly used initialization method that directly takes the result of traditional endmember extraction as the decoder's initial weights, the proposed structure can not only avoid the phenomenon of overfitting and gradient explosion in the iterative process of the network but also can autonomously learn the complex modeling of additive perturbation variability through the powerful fitting ability of the deep network, so as to obtain more robust endmembers.

Therefore, the class-dependent variability autoencoder overall implements the representation of category-related terms in the FLMM model. Note that the values of each element in the matrix Ψ and $\mathbf{dM}$ are initialized as 1 and 0, respectively, which represents the absence of spectral variability. The endmembers are initialized by a traditional spectral endmember extraction algorithm such as VCA [10].

C. Loss Function

The total loss function $\mathcal{L}$ of the proposed SVACA is composed of the following terms:

$$\mathcal{L} = \mathcal{L}_{RE} + \lambda_1 \mathcal{L}_V + \lambda_2 \mathcal{L}_A + \lambda_3 \tag{9}$$

where λ_1, λ_2, and λ_3 are the corresponding regular term coefficients.

1) Reconstruction Loss: To confine pixel-level reconstruction as precisely as feasible, the $\mathcal{L}_{\text{RE}}$ is enforced in the class-dependent variability autoencoder, which is formulated

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TABLE III
NETWORK CONFIGURATION OF CLASS-DEPENDENT VARIABILITY AUTOENCODER IN THE PROPOSED SVACA NETWORK

	Forward Path				Skip Connection			
	Layer	Kernel Size	Filter	Activate Function	Layer	Kernel Size	Filter	Activate Function
Encoder	Conv1	3×3	128	LReLU				
	Conv2	3×3	64	LReLU	ConvSkip	1×1	P	Hardtanh
	Conv3	3×3	P	/				
Hadamard Product								
	Forward Path				Conv4 Parameter Learning			
	Layer	Kernel Size	Filter	Activate Function	Layer		Node	
Decoder	Conv4	1×1	b	/	FC1		32	
					FC2		b	

as

$$\mathcal{L}_{RE} = \mathcal{L}_{\text{LMM}} + \mathcal{L}_{\text{FLMM}} = \|\mathbf{R}' - \mathbf{MA}\|_{\mathbf{F}}^{\mathbf{2}} + \|\mathbf{R}' - (\mathbf{M} + \mathbf{dM})\mathbf{A} \odot \mathbf{\Psi})\|_F^2 \quad (10)$$

where $\mathbf{R}'$ denotes the output of the class-independent variability autoencoder.

2) Variability Loss: The $\mathcal{L}_V$ aims to support the optimization of class-independent variability autoencoder, where the output matrix $\mathbf{R}' = \mathbf{R} - \mathbf{VB}$. Since such variability is distributed evenly in the observed image and its variability amplitude is lower than class-dependent variability, the variability loss can be written as

$$\mathcal{L}_V = \|\mathbf{B}\|_F^2. \quad (11)$$

3) Abundance Loss: To satisfy the abundance nonnegative constraint, the abundance components are limited to the range of 0–1 in the optimization process by the clamp function. Moreover, the sum-to-one constraint is designed as $\mathcal{L}_A$ to limit the sum of each abundance vector to 1 as follows, instead of the commonly used softmax function:

$$\mathcal{L}_A = \sum_{k=1}^{N} |1 - \sum_{p=1}^{P} a_{kp}|_1. \quad (12)$$

Note that this constraint effectively avoids the problem that the softmax function only takes into account the relative probabilities between several materials [51], which leads to a limited convergence accuracy.

4) Endmember Loss: The minimum volume endmember constraint $\mathcal{L}_M$ contributes to maintaining the geometrical structure of endmembers during optimization, which forces the volume of the simplex constructed by endmembers as small as possible. It is defined as

$$\mathcal{L}_M = \frac{1}{2(P-1)!} \det\left(\begin{bmatrix} 1_P^T \\ \mathbf{M}' \end{bmatrix}\right) \quad (13)$$

in which $\mathbf{M}' \in \mathbb{R}^{(P-1)\times P}$ denotes the endmember matrix after reducing the dimension from b to $P-1$.

The pseudo-code of the proposed SVACA network is summarized in Algorithm 1. Note that the class-independent variability autoencoder and the class-dependent variability autoencoder are represented by $f(\cdot)$ and $g(\cdot)$, respectively.

V. EXPERIMENTS

In this section, both synthetic and real-world datasets (namely, Samson and Jasper datasets) are employed to validate the effectiveness of the proposed SVACA. For comparison, several state-of-the-art deep learning-based unmixing networks are adopted for experiments, including CyCU-Net [51], MiSiCNet [52], CNNAEU [53], and DAEU [55].

A. Evaluation Metrics

Two quantitative metrics are adopted to evaluate the performance of endmember extraction and abundance estimation in the experiments. First, to assess the effectiveness of predicted endmembers, the spectral angle distance (SAD) is used. Let $\hat{m}_i$ and m_i be the ith estimated endmember and its corresponding reference spectral signature, respectively. The SAD value (in degrees) between them is defined as

$$\text{SAD}(m_i, \hat{m}_i) = \arccos\left(\frac{\hat{m}_i \cdot m_i}{\|\hat{m}_i\|\|m_i\|}\right) \quad (14)$$

where $\|\cdot\|$ denotes the magnitude of vectors. In general, lower SAD scores mean higher spectral similarity between the compared endmembers.

Second, the abundance estimation accuracy and pixel reconstruction accuracy are evaluated by the root mean squared error (RMSE) between the ground reference vectors and the corresponding estimated vectors, respectively. The definitions are expressed as

$$\text{RMSE}_A = \sqrt{\frac{1}{N}\sum_{j=1}^{N} \|a_j - \hat{a}_j\|_2^2} \quad (15)$$

$$\text{RMSE}_R = \sqrt{\frac{1}{N}\sum_{j=1}^{N} \|r_j - \hat{r}_j\|_2^2}. \quad (16)$$

Note that lower RMSE values mean high consistency with the ground references.

B. Experiments Over Synthetic Dataset

The synthetic data used in the experiments is constructed from DC2 data used in [56], which contains 100 × 100 pixels with 224 spectral bands in the wavelength ranging from 400 to

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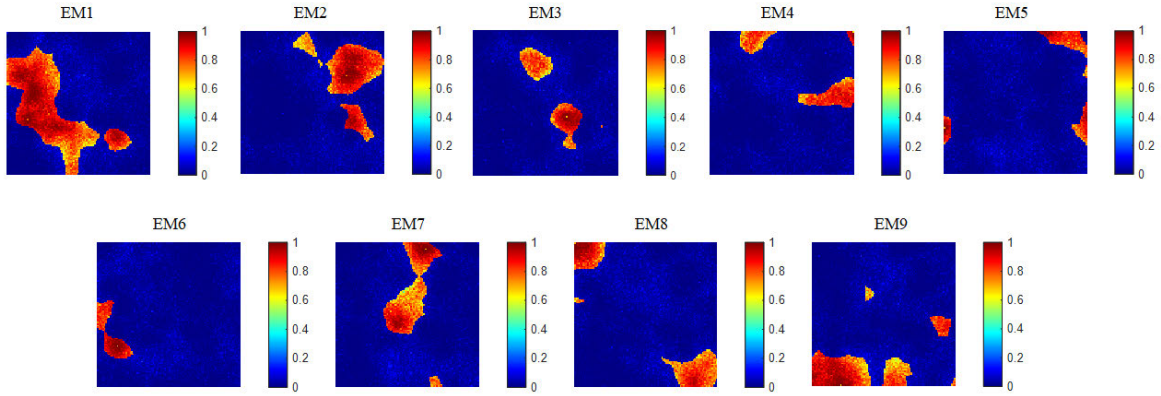


Fig. 2. Ground-truth abundance maps of the synthetic dataset.

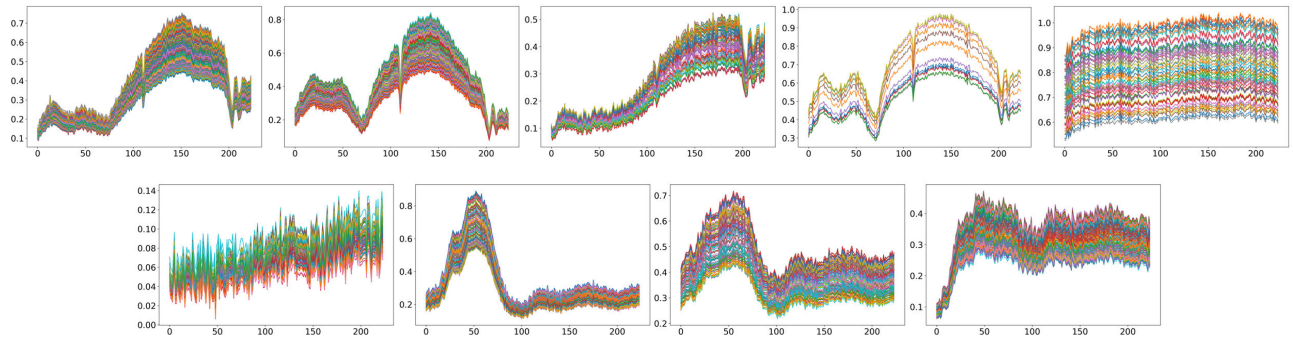


Fig. 3. Endmember clusters after introducing spectral variability over the synthetic dataset.

2500 nm. It is generated by nine endmembers from the USGS spectral library (termed EM1–EM9) under the LMM, and its abundance coefficient follows the Dirichlet distribution and satisfies the ASC and ANC constraints, showing strong spatial correlation. The reference true values for abundance are shown in Fig. 2. To verify the ability to deal with spectral variability, the effects of endmember random scaling, endmember additive perturbation, and abundance random scaling are added. To be specific, the scaling coefficients of endmembers are (0.98, 1.21, 0.95, 1.12, 0.96, 1.19, 0.83, and 0.87), the additive perturbation uses 30 dB Gaussian white noise, and the abundance scaling is multiplied by a random number in the range of (0.75, 1.25) at each endmember of each pixel. Fig. 3 shows the endmember clusters after introducing the above spectral variability over the synthetic dataset.

1) Parameter Analysis: To find the parameters' optimal values, it is first addressed how the proposed SVACA network's parameters affect the unmixing performance. The parameters in the proposed SVACA network include three regularization parameters in the total loss function, i.e., λ_1, λ_2, and λ_3, and the node number of fully connected layers in class-independent variability autoencoder. Fig. 4 depicts the SAD and RMSE results for estimated endmembers and abundances obtained by the proposed SVACA. Note that, λ_1, λ_2, and λ_3 are varied in ($5e^{-6}$, $1e^{-5}$, $5e^{-5}$, $1e^{-4}$), ($5e^{-7}$, $1e^{-6}$, $5e^{-6}$, $1e^{-5}$, $5e^{-5}$), and ($1e^{-5}$, $5e^{-5}$, $1e^{-4}$, $5e^{-4}$, $1e^{-3}$), respectively, and the node number is ranged in (16, 32, 64, 128, 256). It is observed that as the parameter value changes, the performance of endmembers and abundance accuracy share similar trends. In order to achieve relatively optimal unmixing performance, it is suitable for the regularization parameters to take values within ($1e^{-5}$, $1e^{-6}$, $1e^{-4}$). Meanwhile, since the role of the fully connected layer in the class-independent variability autoencoder is related to the learning of the additive perturbation of the endmembers, the endmember accuracy should be taken as the main consideration. Therefore, the node number is suggested to be 64.

TABLE IV
PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS OVER THE SYNTHETIC DATASET

Algorithms		CyCU-Net	MiSiCNet	DAEU	CNNAEU	**SVACA**
SAD	EM1	0.0978	0.1276	0.1786	0.0565	**0.0432**
	EM2	0.0659	0.1065	0.0709	0.0462	**0.0376**
	EM3	0.1029	0.1343	0.0877	0.0796	**0.0485**
	EM4	0.0442	0.0443	0.2400	0.2027	**0.0266**
	EM5	0.0424	0.0160	0.0426	0.0473	**0.0191**
	EM6	0.8566	0.7471	0.1909	**0.1304**	0.1546
	EM7	0.1373	0.1289	**0.0786**	0.1271	0.0825
	EM8	0.1020	0.0764	0.0531	0.2318	**0.0380**
	EM9	0.1455	0.0799	**0.0350**	0.0667	0.0393
mSAD		0.1772	0.1623	0.1086	0.1098	**0.0544**
$RMSE_A$		0.2988	0.1232	0.1259	0.1549	**0.1134**
$RMSE_R$		0.2028	**0.0223**	0.3157	0.2878	0.0287

2) Comparison With State-of-the-Art Methods: Table IV presents the SAD and RMSE results of different unmixing networks over the synthetic dataset, which details the extraction accuracy of each endmember and the mean SAD result of nine endmembers, along with the abundance estimation accuracy and the pixel reconstruction performance. It can be seen that the proposed SVACA network has obtained the optimal extraction accuracy on EM1–EM5 and EM8, and simultaneously achieved the lowest mean SAD value for endmembers and

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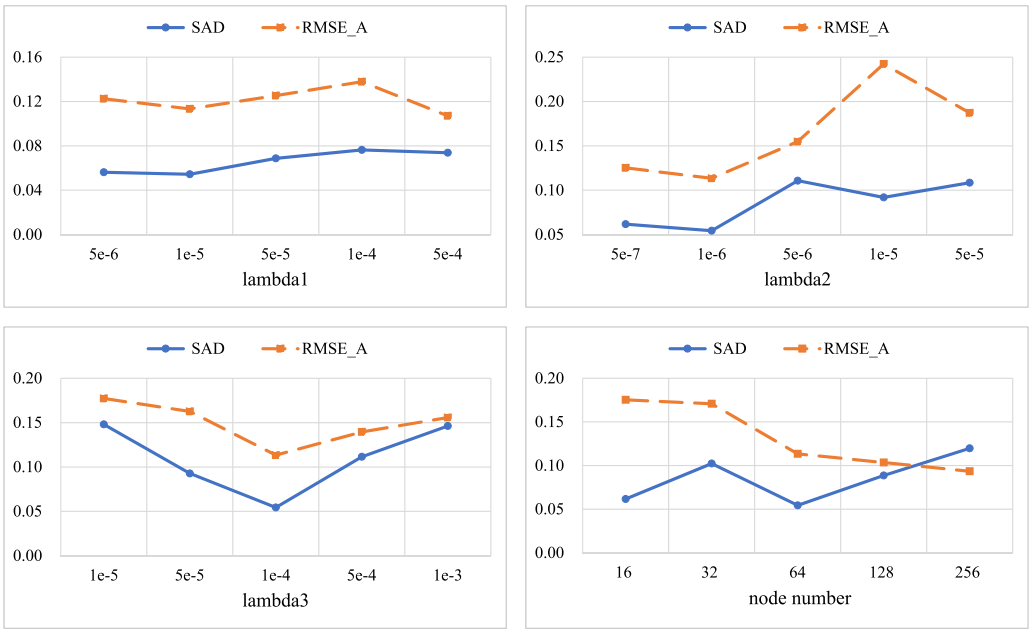


Fig. 4. Parameter analysis of λ_1, λ_2, λ_3, and node number evaluated by SAD and RMSE for both endmember and abundance over synthetic dataset.

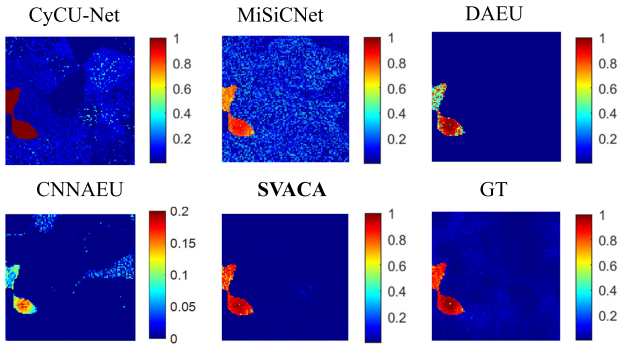


Fig. 5. Abundance maps of the synthetic dataset obtained by the different algorithms, taking EM6 as an example.

TABLE V
PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS UNDER DIFFERENT SNR LEVELS OVER THE SYNTHETIC DATASET

		CyCU-Net	MiSiCNet	DAEU	CNNAEU	SVACA
SNR=10dB	mSAD	0.2182	0.2843	0.2377	0.1508	**0.0665**
	$RMSE_A$	0.3179	0.2238	0.2876	0.2043	**0.1442**
	$RMSE_R$	0.1973	0.1594	0.2783	0.3049	**0.1399**
SNR=15dB	mSAD	0.1988	0.1906	0.1773	0.1440	**0.0635**
	$RMSE_A$	0.3094	0.1573	0.2194	0.2013	**0.1443**
	$RMSE_R$	0.2653	**0.0234**	0.3400	0.3036	0.1246
SNR=20dB	mSAD	0.1906	0.1706	0.1148	0.1365	**0.0634**
	$RMSE_A$	0.2949	0.1294	0.1531	0.2320	**0.1086**
	$RMSE_R$	0.2290	**0.0223**	0.3648	0.3039	0.0638

RMSE result for abundance. In contrast, although the recent CyCU-Net and MiSiCNet can enhance the unmixing ability of the network, their performance affected by spectral variability is falling short of expectations. Moreover, Fig. 5 displays the estimated abundance maps by different methods over the synthetic dataset, where taking EM6 as an example. It can be observed that the proposed SVACA obtains abundance results consistent with the GT. DAEU and CNNAEU are not prepared to well estimate the composition proportion of ground objects. Although MiSiCNet can obtain the abundance estimation of pure ground objects consistent with the true abundances, it also introduces the wrong estimation of the background. These results demonstrate the effectiveness and superiority of the proposed SVACA network in handling various types of spectral variability.

3) Robustness to Noise: Given the reliance on spectral variability, it is crucial to understand whether the proposed method can distinguish between meaningful spectral variation and noise-induced variability. Therefore, three SNR levels (i.e., 10, 15, and 20 dB) are considered over the synthetic dataset to verify the model's robustness to noise. Table V illustrates the average SAD and RMSE results achieved by comparative algorithms. It is observed that the proposed SVACA network achieves the best endmember and abundance estimation performance at all SNR levels, and the best pixel reconstruction performance at the lowest SNR of 10 dB, which is proven to be effective in distinguishing between class-independent noise and class-dependent spectral variability.

C. Experiments Over Samson Dataset

The first real-world dataset, known as Samson dataset,[1] was collected by the SAMSON sensor. It contains 95 × 95 pixels and covers 156 bands and a wavelength range of 401–889 nm, which is free from blank channels or low SNR channels. There are three types of endmembers in the scene, i.e., soil, tree, and water. In the Samson dataset, the regularization parameters are fixed to ($5e^{-4}$, $1e^{-5}$, $1e^{-4}$) after analysis, and the node number of fully connected layers is set to 32.

Table VI lists the quantitative SAD and RMSE values between the ground references and corresponding estimated results obtained by different networks over the Samson dataset. It can be observed that although the DAEU network achieves

[1]Available online at http://lesun.weebly.com/hyperspectral-data-set.html.

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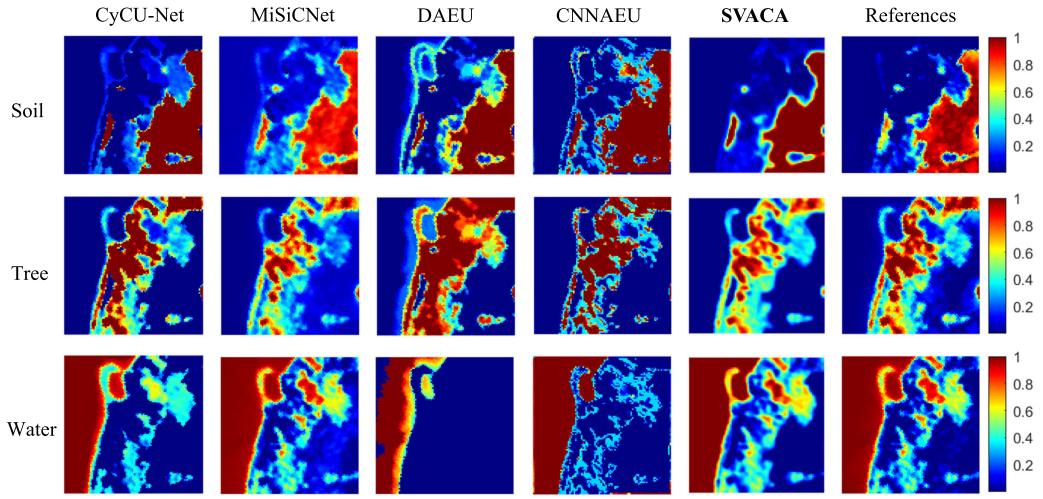


Fig. 6. Abundance maps of the Samson dataset obtained by the different algorithms.

TABLE VI
PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS OVER THE SAMSON DATASET

Algorithms		CyCU-Net	MiSiCNet	DAEU	CNNAEU	**SVACA**
SAD	Soil	2.8922	1.7769	2.6586	2.2237	**0.5842**
	Tree	2.2200	4.0767	3.5581	2.3766	**1.7923**
	Water	8.9480	12.0466	**2.7851**	6.0626	3.7848
mSAD		4.6867	5.9645	3.0023	3.5523	**2.0512**
$RMSE_A$		0.1914	0.0839	0.2966	0.2478	**0.0747**
$RMSE_R$		0.0529	0.0230	0.3710	0.1328	**0.0225**

TABLE VII
PERFORMANCE OBTAINED BY DIFFERENT ALGORITHMS OVER THE JASPER DATASET

Algorithms		CyCU-Net	MiSiCNet	DAEU	CNNAEU	**SVACA**
SAD	Tree	**1.6878**	2.4829	5.2133	4.1680	4.5235
	Water	12.6471	16.5789	**1.5884**	3.5273	7.2486
	Soil	3.6104	10.0667	**1.6768**	8.3826	3.1914
	Road	7.4693	9.3596	9.2556	7.2014	**3.1729**
mSAD		6.3541	9.6200	**4.4347**	5.8213	4.5341
$RMSE_A$		0.1162	0.2574	0.1372	0.3063	**0.1128**
$RMSE_R$		0.0439	0.0263	0.2825	0.2226	**0.0222**

the optimal endmember estimation of water endmember, the proposed SVACA ranks first compared to other state-of-the-art algorithms in terms of the overall endmember extraction, abundance estimation, and pixel reconstruction.

Moreover, for qualitative evaluation, Fig. 6 depicts the abundance maps estimated by different unmixing networks and their corresponding ground references. As shown in Fig. 6, the proposed SVACA network obtains the most similar abundance maps compared to the ground references, especially for soil endmember, while the abundance maps obtained by comparative methods all show confounding effects with other ground objects in the background. Overall, the above results show that the proposed SVACA network offers excellent unmixing performance even on real-scene datasets, which verifies the superiority and effectiveness of the FLMM and SVACA network structure design.

D. Experiments Over Jasper Dataset

The second real-world dataset, i.e., Jasper dataset, was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor,[2] which spans 224 bands and covers the wavelengths from 380 to 2500 nm. A 100 × 100 pixels cropped image was used in this experiment, which contains 198 reflectance bands selected after removing the water-vapor absorption bands and noisy bands. There are four main endmembers in this scene, which are identified as trees, water, soil, and road. Similarly, the regularization parameters in the Jasper dataset experiment are the same as the synthetic experiment settings, but the node number of the fully connected layer is set to 32.

Table VII and Fig. 7 show the quantitative and qualitative evaluation results by different algorithms over the Jasper dataset, respectively. As shown in Table VII, the proposed SVACA realized the suboptimal extraction of overall endmembers and the optimal estimation of abundances, thus achieving the best reconstruction of HSIs. Meanwhile, it is observed that in the comparative methods, it is difficult to obtain an optimal estimate of both endmember extraction and abundance estimation. Sometimes, an optimal endmember extraction performance may be realized at the expense of abundance accuracy.

Moreover, the visual abundance assessment is displayed in Fig. 7. Compared to the ground references, most comparison methods failed to completely separate the road endmember, especially the MiSiCNet, DAEU, and CNNAEU. However, the abundance maps obtained by the proposed SVACA present high consistency to the ground references of all four materials, which benefits from the modeling and learning of class-dependent and -independent spectral variability.

E. Ablation Study

The ablation experiments over two real-world datasets are carried out to assess the effectiveness of each component within the proposed SVACA. The main components of the

[2]Available online at https://rslab.ut.ac.ir/data.

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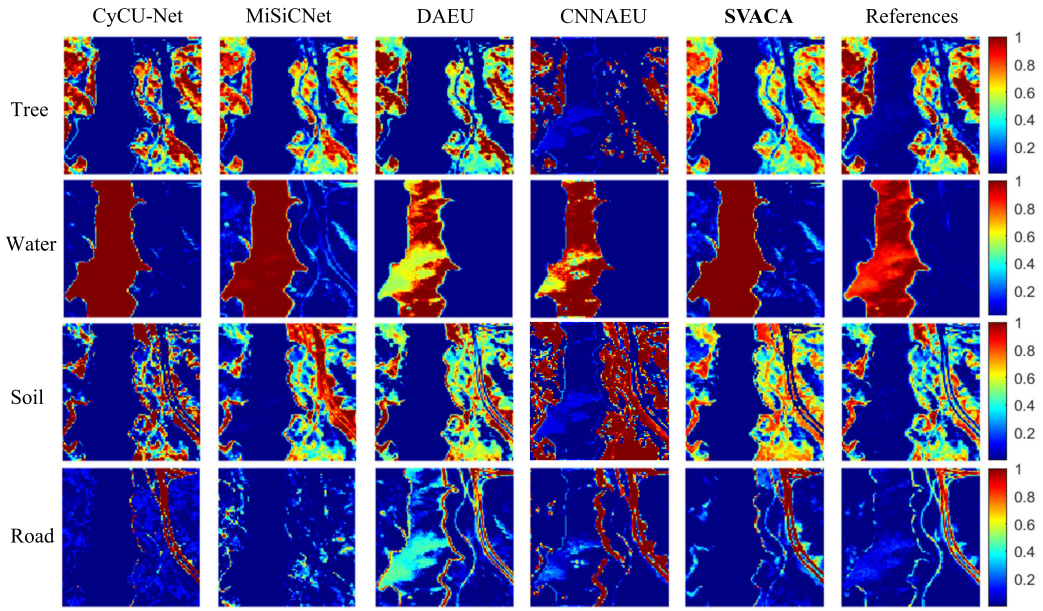


Fig. 7. Abundance maps of the Jasper dataset obtained by the different algorithms.

TABLE VIII
ABLATION EXPERIMENTS OF THE PROPOSED SVACA CONDUCTED OVER SAMSON AND JASPER DATASETS

Models	Class-independent Variability	Class-dependent Scaling Variability	Class-dependent Additional Variability	Samson data set		Jasper data set	
				SAD	$RMSE_A$	SAD	$RMSE_A$
M_1	√			5.0191	0.0952	5.6895	0.1211
M_2		√		4.8255	0.0837	6.3598	0.1184
M_3			√	2.8877	0.1563	5.3973	0.1332
M_4	√	√		4.2743	0.0925	6.1421	0.1194
M_5	√		√	3.9935	0.1454	5.6952	0.1209
M_6		√	√	3.4223	0.0869	4.7263	0.1172
M_7(SVACA)	√	√	√	**2.0512**	**0.0747**	**4.5341**	**0.1128**

SVACA include the class-independent variability autoencoder and class-dependent variability autoencoder, where the latter can be divided into an additional perturbation module and a multiplicative scaling module. By varying the combination of these three spectral variability modules, six degraded variants of the proposed SVACA can be denoted by M_1–M_6. The detailed configuration and comparison results of these six variants are shown in Table VIII, where mean SAD and RMSE results for endmembers and abundances are adopted to evaluate the unmixing performance.

According to Table VIII, only employing one type of spectral variability in M_1, M_2, and M_3 cannot achieve satisfied unmixing results, neither in the endmember extraction accuracy nor in the abundance estimation performance. In addition, the model equipped with class-independent variability and one type of class-dependent variability in M_4 and M_5 reflects opposite influences on the accuracy of endmembers and abundances. On the other hand, when fully considering class-dependent variability, the model M_6 gains greatly improved performance, which indicates that the modeling of class-dependent variability plays a key role in the unmixing accuracy. Finally, the proposed SVACA model achieves systematic modeling of spectral variability by introducing one type of class-independent spectral variability and two types of class-dependent spectral variability and obtains superior performance in terms of endmember and abundance estimation.

VI. CONCLUSION

In this article, a novel FLMM and an SVACA are proposed to deal with the spectral variability within HSIs. Unlike the existing mixture models that only identify certain spectral variability, the proposed FLMM model can accommodate both class-dependent and class-independent spectral variability in the absence of prior information. And the learning and representation of both endmembers and spectral variability are further realized by the proposed SVACA network. Experimental results over synthetic and real-world datasets validate that the proposed SVACA outperforms state-of-the-art unmixing algorithms when encountering complex and diverse spectral variability. However, the proposed method has some limitations that it only considers the characterization of spectral variability in the linear mixing mode of ground objects, and that in the more complex but closer to the actual scene nonlinear mixing mode remains to be studied.

REFERENCES

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] Y. Zhong, X. Wang, S. Wang, and L. Zhang, "Advances in spaceborne hyperspectral remote sensing in China," *Geo-Spatial Inf. Sci.*, vol. 24, no. 1, pp. 95–120, Jan. 2021.

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[3] M. Ma, F. Li, Y. Hu, Z. Wang, and S. Mei, "Joint spatial and spectral graph-based consistent self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520616.

[4] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[5] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.

[6] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[7] M. Li, B. Yang, and B. Wang, "EMLM-net: An extended multilinear mixing model-inspired dual-stream network for unsupervised nonlinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509116.

[8] Y. Zhang, J. Tian, T. Zhao, and S. Mei, "Spatial resolution enhancement of remote sensing hyperspectral images with localized spatial–spectral dictionary pair," *IEEE Access*, vol. 8, pp. 61051–61069, 2020.

[9] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–277, Oct. 1999.

[10] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[11] X.-R. Feng, H.-C. Li, R. Wang, Q. Du, X. Jia, and A. Plaza, "Hyperspectral unmixing based on nonnegative matrix factorization: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4414–4436, 2022.

[12] X. Li, X. Zhang, Y. Yuan, and Y. Dong, "Adaptive relationship preserving sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504516.

[13] C. Cui, X. Wang, S. Wang, L. Zhang, and Y. Zhong, "Unrolling nonnegative matrix factorization with group sparsity for blind hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 31, 2023, Art. no. 5516712.

[14] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A blind multiscale spatial regularization framework for kernel-based spectral unmixing," *IEEE Trans. Image Process.*, vol. 29, pp. 4965–4979, 2020.

[15] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[16] L. Ren, D. Hong, L. Gao, X. Sun, M. Huang, and J. Chanussot, "Hyperspectral sparse unmixing via nonconvex shrinkage penalties," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500415.

[17] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.

[18] R. A. Borsoi, T. Imbiriba, and P. Closas, "Dynamical hyperspectral unmixing with variational recurrent neural networks," *IEEE Trans. Image Process.*, vol. 32, pp. 2279–2294, 2023.

[19] Y. Su et al., "DAAN: A deep autoencoder-based augmented network for blind multilinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512715.

[20] L. Qi, Z. Chen, F. Gao, J. Dong, X. Gao, and Q. Du, "Multiview spatial–spectral two-stream network for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502016.

[21] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.

[22] J. Theiler, A. Ziemann, S. Matteoli, and M. Diani, "Spectral variability of remotely sensed target materials: Causes, models, and strategies for mitigation and robust exploitation," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 8–30, Jun. 2019.

[23] W. Gao, J. Yang, and J. Chen, "Proportional perturbation model for hyperspectral unmixing accounting for endmember variability," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[24] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[25] L. Drumetz, J. Chanussot, C. Jutten, W.-K. Ma, and A. Iwasaki, "Spectral variability aware blind hyperspectral image unmixing based on convex geometry," *IEEE Trans. Image Process.*, vol. 29, pp. 4568–4582, 2020.

[26] T. Uezato, M. Fauvel, and N. Dobigeon, "Hyperspectral unmixing with spectral variability using adaptive bundles and double sparsity," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3980–3992, Jun. 2019.

[27] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "A data dependent multiscale model for hyperspectral unmixing with spectral variability," *IEEE Trans. Image Process.*, vol. 29, pp. 3638–3651, 2020.

[28] Y. Zhou, A. Rangarajan, and P. D. Gader, "A Gaussian mixture model representation of endmember variability in hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2242–2256, May 2018.

[29] A. Zouaoui, G. Muhawenayo, B. Rasti, J. Chanussot, and J. Mairal, "Entropic descent archetypal analysis for blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 32, pp. 4649–4663, 2023.

[30] S. S. Vijayashekhar, V. S. Deshpande, and J. S. Bhatt, "A practical approach for hyperspectral unmixing using deep learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] S. S. Vijayashekhar and J. S. Bhatt, "A blind spectral unmixing in wavelet domain," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10287–10302, 2021.

[32] P.-A. Thouvenin, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral unmixing with spectral variability using a perturbed linear mixing model," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 525–538, Feb. 2016.

[33] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.

[34] T. Imbiriba, R. A. Borsoi, and J. C. M. Bermudez, "Generalized linear mixing model accounting for endmember variability," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2018, pp. 1862–1866.

[35] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[36] G. Zhang et al., "Spectral variability augmented sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527413.

[37] P. Lukeš, P. Stenberg, M. Rautiainen, M. Mõttus, and K. M. Vanhatalo, "Optical properties of leaves and needles for boreal tree species in Europe," *Remote Sens. Lett.*, vol. 4, no. 7, pp. 667–676, Jul. 2013.

[38] B. Wang, S. Zhang, Y. Feng, S. Mei, S. Jia, and Q. Du, "Hyperspectral imagery spatial super-resolution using generative adversarial network," *IEEE Trans. Comput. Imag.*, vol. 7, pp. 948–960, 2021.

[39] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.

[40] Y. Su, A. Marinoni, J. Li, J. Plaza, and P. Gamba, "Stacked nonnegative sparse autoencoders for robust hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1427–1431, Sep. 2018.

[41] S. Ozkan, B. Kaya, and G. B. Akar, "EndNet: Sparse autoencoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.

[42] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.

[43] P. Jia, M. Zhang, and Y. Shen, "Deep spectral unmixing framework via 3D denoising convolutional autoencoder," *IET Image Process.*, vol. 15, no. 7, pp. 1399–1409, May 2021.

[44] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[45] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Deep generative endmember modeling: An application to unsupervised spectral unmixing," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 374–384, 2020.

[46] Q. Jin, Y. Ma, F. Fan, J. Huang, X. Mei, and J. Ma, "Adversarial autoencoder network for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4555–4569, Aug. 2021.

[47] A. Min, Z. Guo, H. Li, and J. Peng, "JMnet: Joint metric neural network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505412.

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[48] X. Zhang, Y. Sun, J. Zhang, P. Wu, and L. Jiao, "Hyperspectral unmixing via deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1755–1759, Nov. 2018.
[49] B. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Spectral–spatial hyperspectral unmixing using multitask learning," *IEEE Access*, vol. 7, pp. 148861–148872, 2019.
[50] Z. Hua, X. Li, Q. Qiu, and L. Zhao, "Autoencoder network for hyperspectral unmixing with adaptive abundance smoothing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1640–1644, Sep. 2021.
[51] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.
[52] B. Rasti, B. Koirala, P. Scheunders, and J. Chanussot, "MiSiCNet: Minimum simplex convolutional network for deep hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522815.
[53] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2020.
[54] B. Rasti, B. Koirala, P. Scheunders, and P. Ghamisi, "UnDIP: Hyperspectral unmixing using deep image prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504615.
[55] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.
[56] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A fast multiscale spatial regularization for sparse hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 598–602, Apr. 2019.



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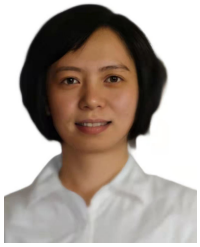
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